

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

Headline finding:

50% task-completion time horizon doubles every ~207 days (~7 months)

Audience: AI safety researchers, frontier AI lab leadership, policy analysts

Source: METR empirical benchmark study

A New Metric: 50% Task-Completion Time Horizon

Methodology overcomes benchmark saturation by measuring capability against task duration

1) Build diverse suite of 170 software tasks

Tasks span 3 seconds to 8 hours, sourced from HCAST, RE-Bench, and SWAA.



2) Measure humans and model success

Skilled humans time each task; models attempt with pass/fail scoring to estimate success probabilities.



3) Fit logistic model → find 50% horizon

Estimate task duration where model succeeds 50% of the time; robust trend fit reaches $R^2 = 0.97$.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Task-suite composition and examples across increasing duration

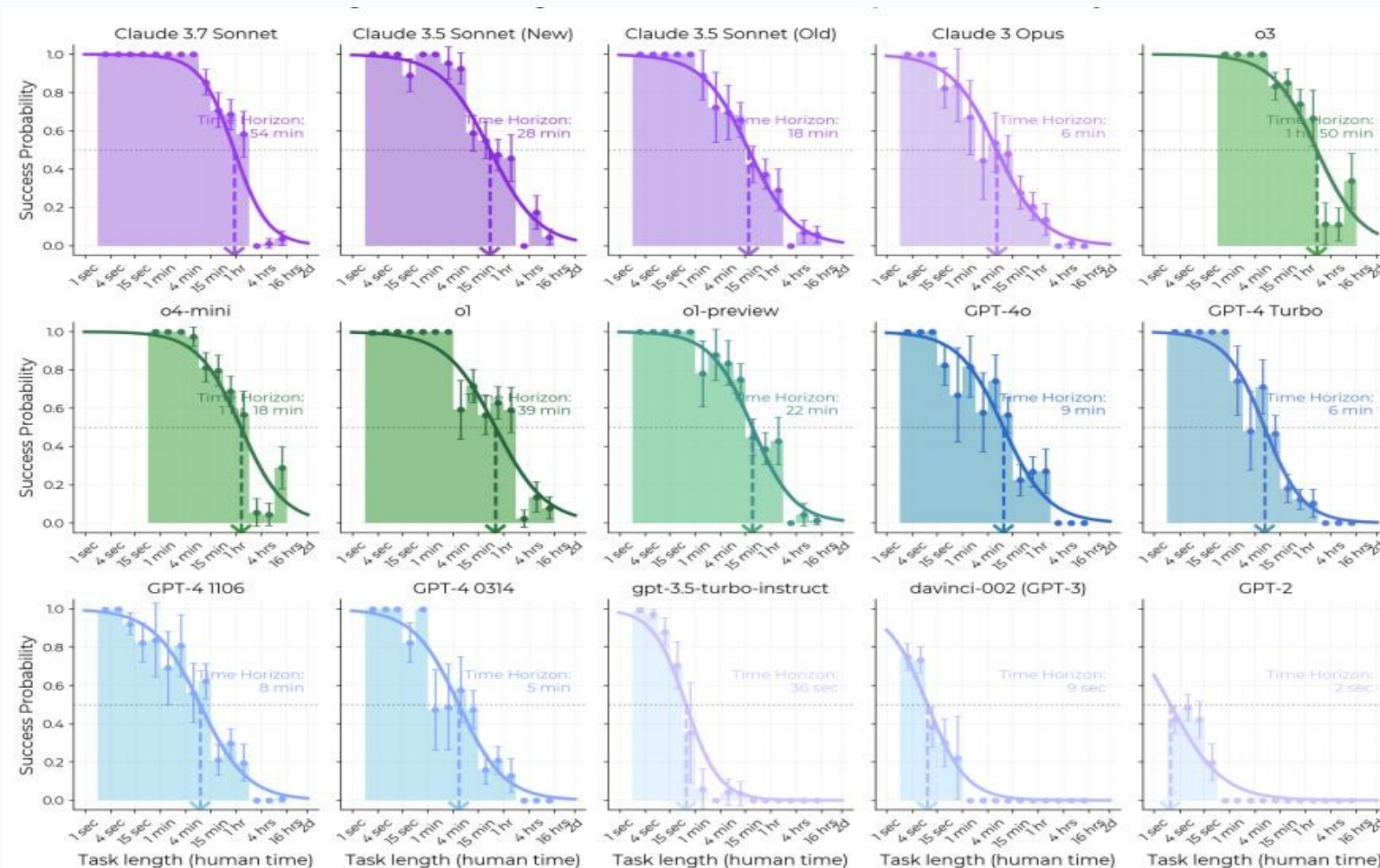
Structured Task Suite Overview

Source	Task Count	Duration Range	Example Tasks
HCAST	97	3 seconds to 1 hour+	find_shell_script (3s); optimize_sorting (12m); debug_flask_api (48m)
RE-Bench	7	8 minutes to 8 hours	docker_compose_fix (8m); refactor_legacy_module (2h); cuda_backtesting (8h)
SWAA	66	15 seconds to 4 hours+	regex_cleanup (15s); write_ci_pipeline (35m); distributed_cache_patch (4h)

Suite total: 170 tasks covering micro-actions to multi-hour engineering workflows.

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years

Exponential growth in 50% completion time horizon across model releases



Trend Parameters

Doubling time: ~207 days
(~7 months)

Goodness of fit:
 $R^2 = 0.97$

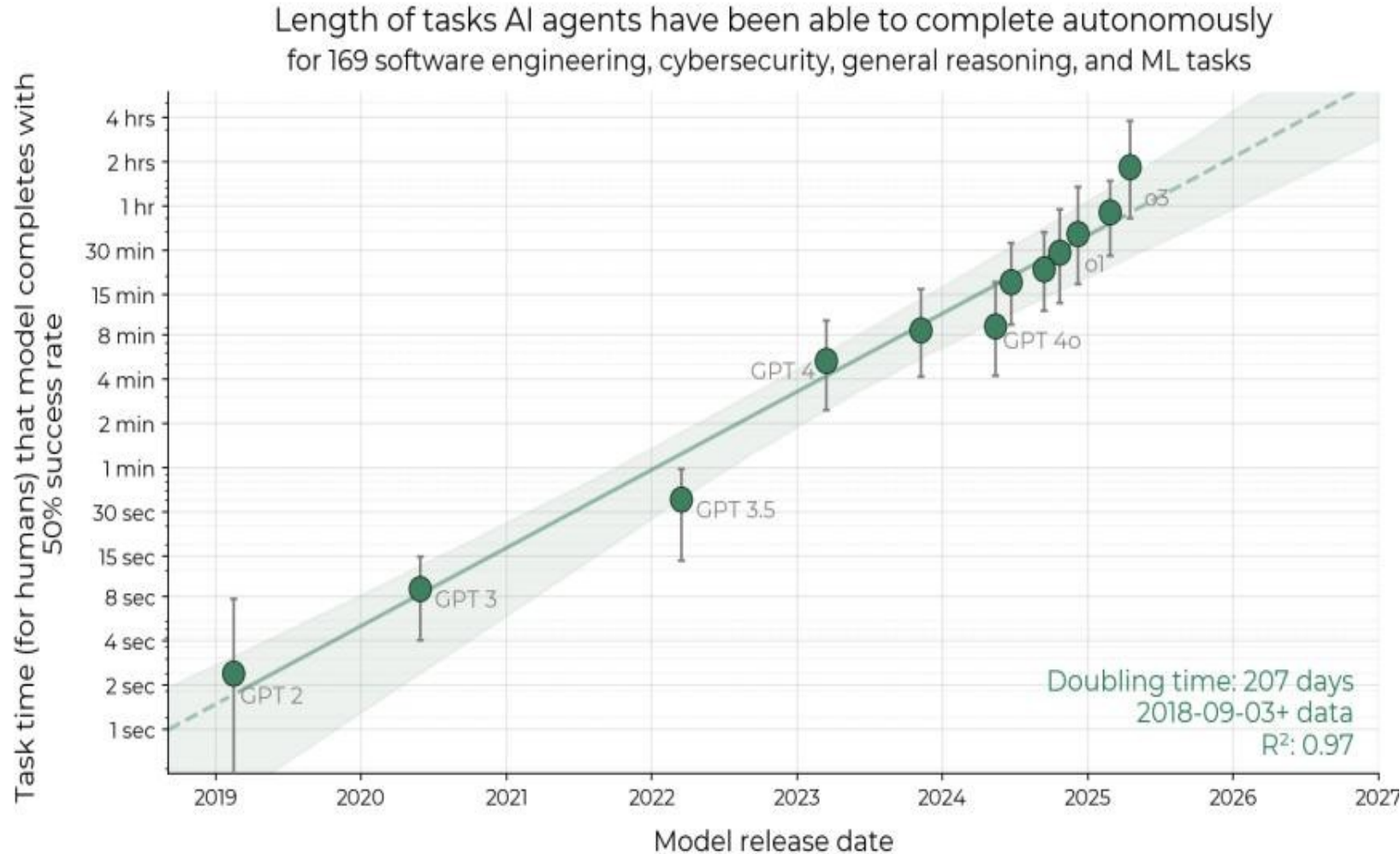
Milestones:

- GPT-2: ~2 sec
- GPT-4 (0314): ~5 min
- o3: ~1h 50m

**2019→2025: ≈3,500× horizon
increase**

Success Probability Drops Sharply as Task Length Increases for All Models

Frontier models shift curves rightward, extending feasible autonomous task duration



Interpretation

- Each curve: success rate vs. task time
- Longer tasks reduce success for all models
- Newer models move the curve right
- Frontier reference points:
 - Claude 3.7 Sonnet: ~54 min
 - o3: ~1h 50m
- Older baseline: GPT-2 ~2 sec

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Capability gains are substantial but uneven across task structure

Improved logical reasoning

Models sustain multi-step decomposition and maintain coherent plans over longer trajectories.

Better tool use

Higher reliability in coding tools, shell operations, and iterative verification loops.

Reliability + self-awareness

More consistent execution quality and better recognition of uncertainty/failure modes.

Adaptation to mistakes

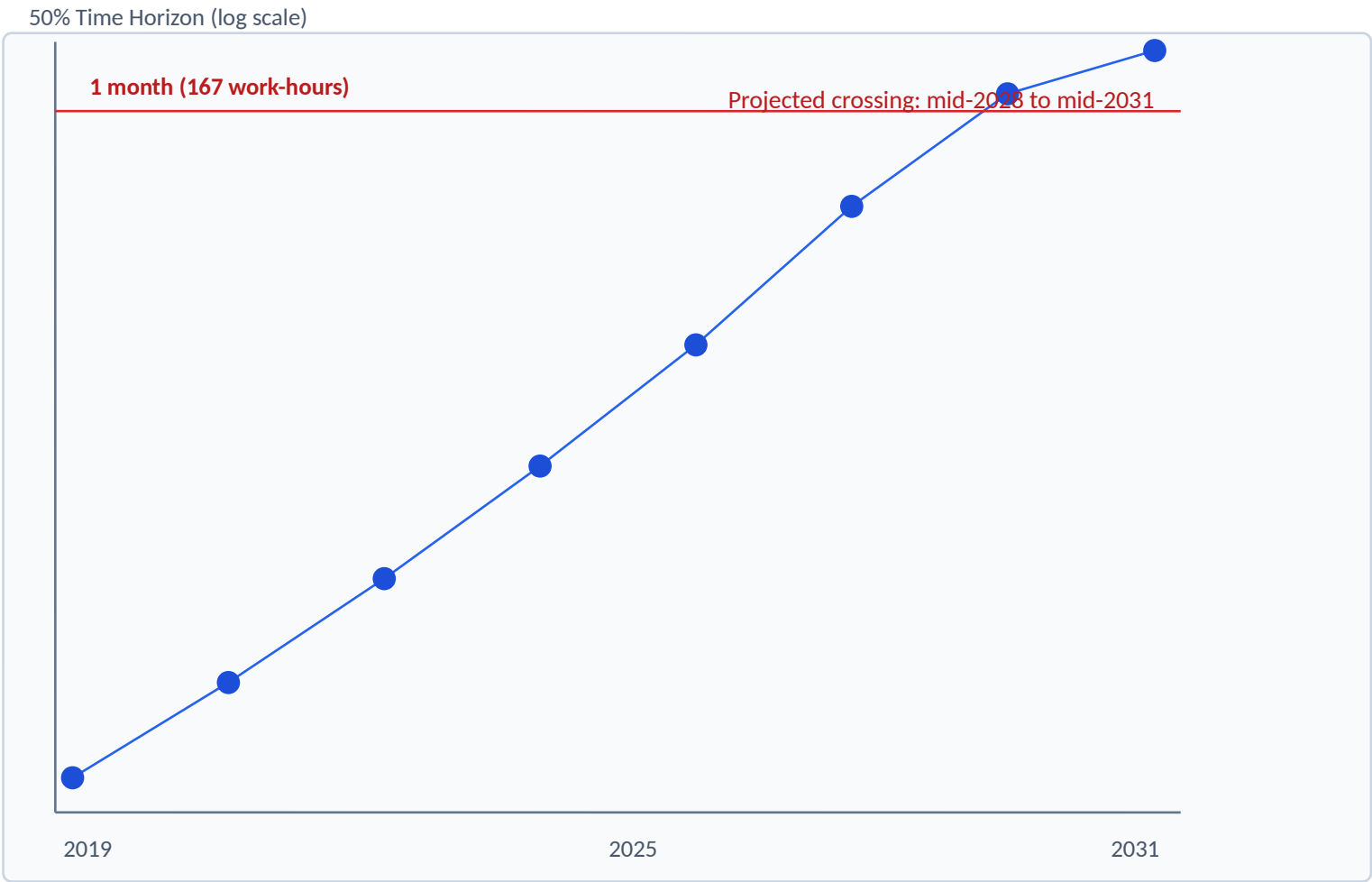
Improved recovery after dead ends enables completion of longer engineering workflows.

Limitation: performance is markedly lower on less structured, "messier" tasks.

External validity remains open: benchmark tasks may not fully represent the full distribution of real-world intellectual labor, even though supplementary analyses found little evidence of slower trends on more realistic tasks.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

Forward-looking implications under continuation of current trend



Forecast Notes

- Naive exponential extrapolation only
- 2023–2025 growth appears ~20% faster than 2019–2025 average
- Potential acceleration is plausible but uncertain
- Trend breaks and deployment constraints could shift outcomes

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety

Governance

Implications for frontier labs, evaluators, and policy institutions

- 1 50% time horizon is a robust, intuitive autonomy metric ($R^2 = 0.97$ over 6 years).
- 2 Horizon doubles every ~7 months, with possible post-2024 acceleration.
- 3 Current frontier (o3) reaches ~2 hours of skilled human-equivalent software work.
- 4 External validity is unresolved: benchmark coverage may not mirror real-world job distributions.
- 5 **Safety governance should anticipate rapid extension from hour-scale to month-scale autonomy.**

Bottom line: capability growth is fast enough that governance, evals, and safeguards must scale ahead of deployment timelines.